Lab Report: Text classification with neural networks

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 80 percent training data and the rest for testing. All experiments were run on a single GPU with 24 GB of memory.

2 Results

Figure 1 shows the accuracy of CNN by dataset.

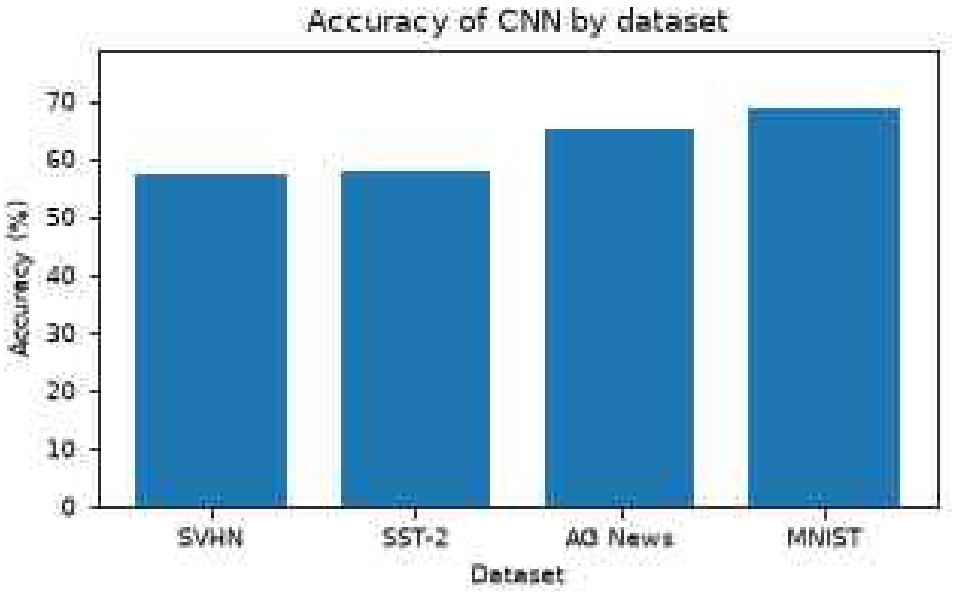

Figure 1: Accuracy of CNN by dataset for the models in this experiment.

On SST-2, CNN obtains a accuracy of 58.0%. On SST-2, CNN obtains a accuracy of about 60%.

Figure 2 shows the accuracy by dataset.

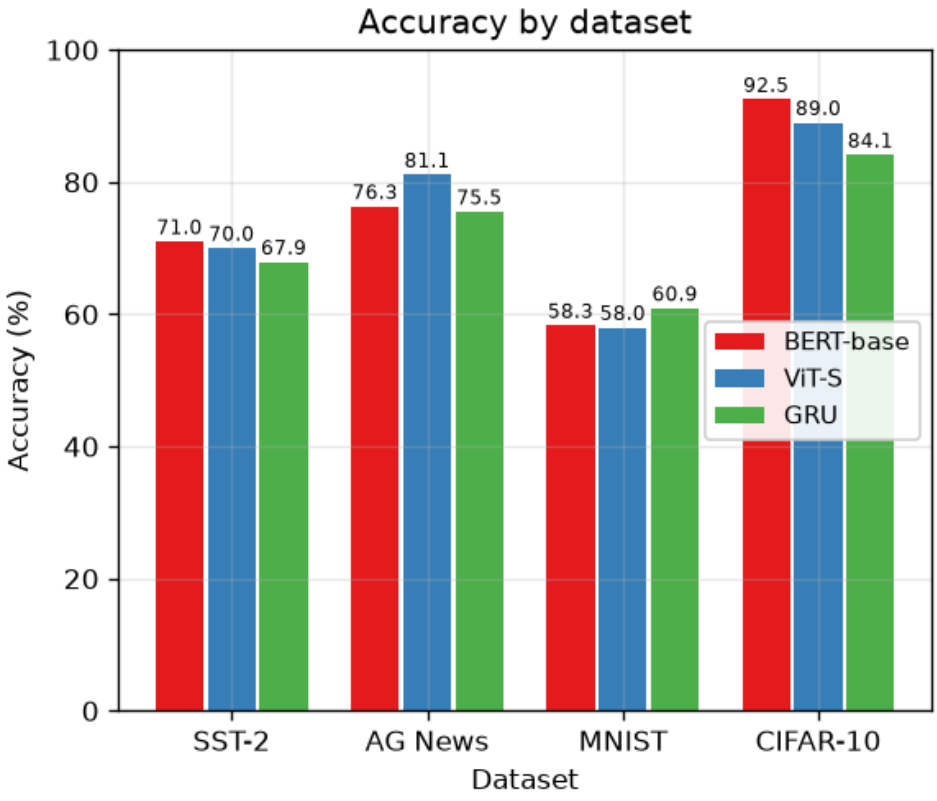

Figure 2: Accuracy by dataset for the models in this experiment.

BERT-base achieves 81.1% accuracy on AG News. On CIFAR-10, ViT-S obtains a accuracy of 139.0%. As shown in Figure 2, on CIFAR-10, BERT-base obtains a accuracy of 92%. BERT-base reaches 68.3% accuracy on MNIST.

3 Discussion

We note that the training was stable for all models.